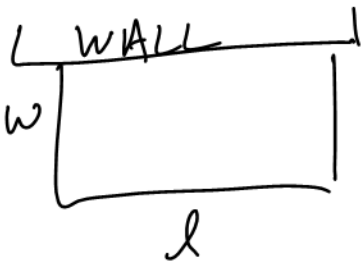
Goal: Solve optimization problems involving cost, revenue, and profit.

Cost Functions

Factoring in costs can be incorporated in a variety of ways. For example, it could act as an objective function:

Example 1. If the cost of fencing material is \$55/ft, what is the minimum cost to produce an enclosed space with an area of 490 000ft^2.



similar to the diagram below

$$\text{Cost} = \left[\frac{\text{Price}}{\text{ft. of fencing}}\right]\left[\text{amount of ft. used}\right]$$

$$= 55(2w + l)$$

Obj: min $C(w,l) = 55(2w+l)$

Given: $lw = 490\,000$

$$l = \frac{490\,000}{w}$$

$\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$

$$C(w) = 55\left(2w + \frac{490000}{w}\right)$$

$$C(w) = 110w + \frac{26\,950\,000}{w}$$

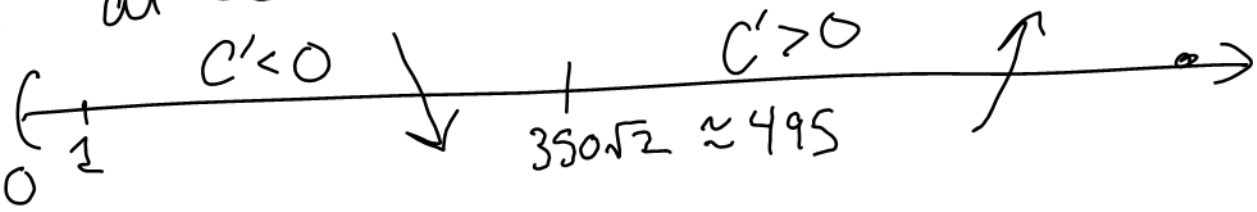
$$\frac{dC}{dw} = 110 - \frac{26\,950\,000}{w^2} \overset{\text{set}}{=} 0$$

$$110 = \frac{26950000}{w^2}$$

$$w^2 = 245000$$

$$w = 350\sqrt{2} \approx$$

Only 1 Crit Num + Rel. Min at w $\Longrightarrow$ Abs. Min at w



Find the minimum cost:

$$C(w) = 110w + \frac{26\,950000}{w}$$

$$C(350\sqrt{2}) = 77000\sqrt{2}$$

$$\approx \$108,894.44$$

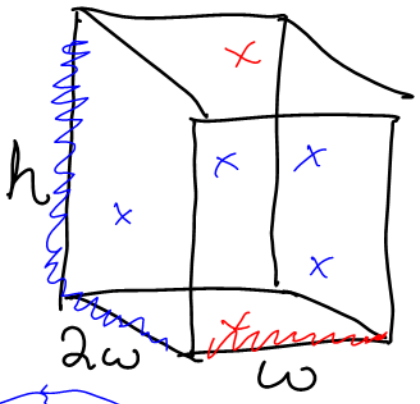
Example 2. Jessica wants to make a box such that:

(i) The volume is 144ft^3;

(ii) The length is double the width;

(iii) The top and bottom are made out of metal;

(iv) The sides are made out of wood.



If the cost of wood is $\$10/\text{ft}^2$ and the cost of metal is $\$20/\text{ft}^2$, what should the dimensions be to make the box as cheaply as possible?

Determine our obj:

$$\text{Cost} = \left[\begin{array}{c}\text{Cost of}\\ \text{Wood}\end{array}\right] + \left[\begin{array}{c}\text{Cost of}\\ \text{metal}\end{array}\right]$$

$$= \left[\frac{\text{cost}}{\text{sq.ft}}\right]\left[\begin{array}{c}\text{Sq.ft}\\ \text{of wood}\end{array}\right] + \left[\frac{\text{cost}}{\text{sq.ft}}\right]\left[\begin{array}{c}\text{sq. ft.}\\ \text{of metal}\\ \text{used}\end{array}\right]$$

$$= 10\left(2wh + 2(2wh)\right) + 20(2\cdot 2w^2)$$

Obj: Min $C(w,h) = 60hw + 80w^2$

Given: $144 = 2w^2h$

$$h = \frac{144}{2w^2} = \frac{72}{w^2}$$

$$C(w) = 60\left(\frac{72}{w^2}\right)w + 80w^2$$

$$C(w) = \frac{4320}{w} + 80w^2$$

$$\frac{dC}{dw} = -\frac{4320}{w^2} + 160w \overset{\text{set}}{=} 0$$

$$160w = \frac{4320}{w^2}$$

$$w^3 = \frac{4320}{160}$$

$$w^3 = 27$$

$$w = 3$$

We need to find the width, length, and height

Width : 3

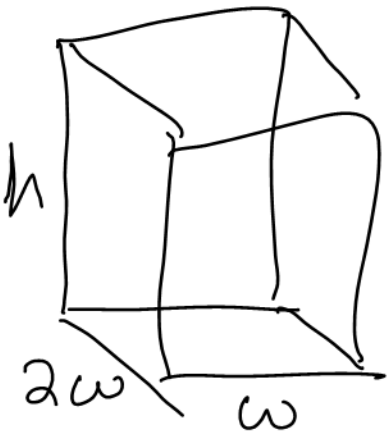
Length : $2(3) = 6$

Height : $h = \frac{72}{w^2} \Rightarrow h = \frac{72}{3^2} = 8$

$3' \times 6' \times 8'$

Costs can also act as a constraint:

Example 3. Using the same setup as **Example 2.**, what is the largest box we can make if we have a budget of \$240?



Obj: max $V_{(w,h)} = 2w^2h$

Given: $\begin{bmatrix}\text{Cost to}\\ \text{produce}\end{bmatrix} = 240$

$60hw + 80w^2 = 240$

$60hw = 240 - 80w^2$

$h = \frac{240 - 80w^2}{60w}$

$h = \frac{4}{w} - \frac{4}{3}w$

$V(w) = 2w^2\left(\frac{4}{w} - \frac{4}{3}w\right)$

$V(w) = 8w - \frac{8}{3}w^3$

$\frac{dV}{dw} = 8 - 8w^2 \overset{\text{set}}{=} 0$

$8w^2 = 8$

$w^2 = 1$

$w = 1$

Find max volume:

$V(w) = 8w - \frac{8}{3}w^3$

$V(1) = 8 - \frac{8}{3} = \frac{16}{3} = 5\frac{1}{3}$ ft^3

h

Dimensions: $1' \times 2' \times \frac{8}{3}'$

Revenue and Profit

Revenue: The total money earned. In symbols,

$$\text{Revenue} = \left[\frac{\text{Price}}{\text{Unit}}\right]\left[\#\text{ of units sold}\right]$$

$$R(p,q) = pq$$

Example 4. A company's marketing department says that the number of units sold starts at 720, then decreases by 15 for every dollar increase in price.

(a) What should the price be to maximize revenue?

Obj: max $R(p,q) = pq$

Given: $q = 720 - 15p$

$$R(p) = p(720 - 15p)$$

$$R(p) = 720p - 15p^2$$

$$\frac{dR}{dp} = 720 - 30p \quad \text{set } 0$$

$$720 = 30p$$

$$p = 24$$

Profit: The total money earned after dealing with costs. In symbols,

Profit = Revenue − Costs

= [Price / Unit] [# of units sold] − [Cost / Unit] [# of units sold]

= p q − C q

$$R(p,q) = pq - Cq$$

(b) What should the price be to maximize profit? Provided the cost is \$12/unit

Obj: Max $R(p,q) = pq - 12q$

Given: $q = 720 - 15p$

$$R(p) = p(720-15p) - 12(720-15p)$$

$$R(p) = 720p - 15p^2 - 8640 + 180p$$

$$R(p) = -15p^2 + 900p - 8640$$

$$\frac{dR}{dp} = -30p + 900 \overset{\text{set}}{=} 0$$

$$30p = 900$$

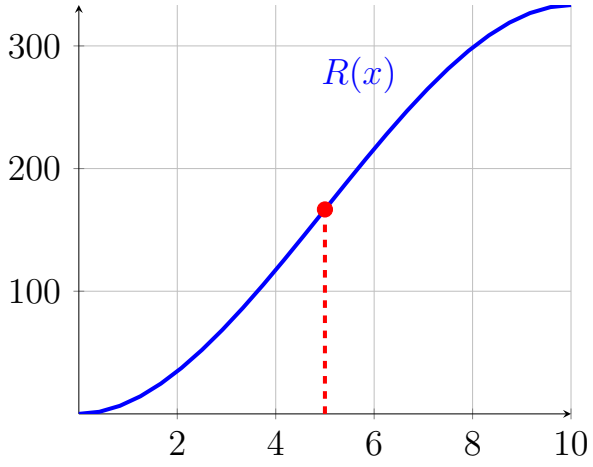
$$p = 30$$

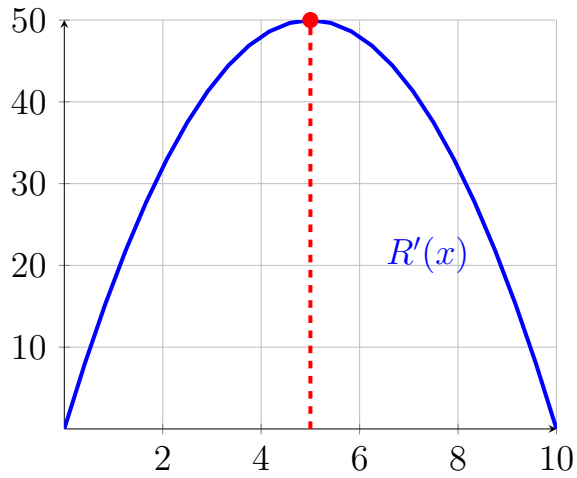
Therefore, when factoring in costs, the price rose from \$24 to \$30.

NOTE: The remainder of this section is NON-EXAMINABLE.

The Point of Diminishing Returns

Definition. When the marginal cost/revenue/profit obtains an absolute maximum, this is called the **point of diminishing returns**.





Example 5. A company uses the following function:

$$R(x) = 10x^2 - \frac{2}{3}x^3; \quad 0 \le x \le 10$$

to model revenue after spending x million dollars on advertising. Find and interpret the point of diminishing returns.

Obj: max $\frac{dR}{dx} = 20x - 2x^2$

Given: $x \in [0,10]$

$$\frac{dR}{dx} = 20x - 2x^2$$

$$\frac{d}{dx}\left(\frac{dR}{dx}\right) = 20 - 4x \overset{\text{set}}{=} 0$$

$$20 = 4x$$

$$x = 5$$

x	0	5	10
$20x - 2x^2 = \frac{dR}{dx}$	0	50	0

abs. max at $x = 5$

Conclusion: The point of diminishing returns is a \$5 million Ie, at \$5 million, the rate the revenue increases by spending more on ad. Sows